

INFO1 - Complete RAG Knowledge Base

Agri-Companies Pitch Manufacturing Service FAB

PRODUCTS & SERVICES

Your Ultimate Bio-input Manufacturer

Quickly Add Microbials To Portfolio @Zero R&D Costs

- Diversify your Product Portfolio without R&D Costs and Efforts
- Manage Geopolitical Uncertainties and Supply Chain Shocks Without Sacrificing Revenues
- Claim Premium Margins and Boost Bottomline
- Participate in Integrated Nutrient/Pest Management Practices

Patented Formulation Technology

IG4 Technology (Indian Patent No.: 421141)

FIB-SOL's novel gel-based formulations have:

- Industry-leading microbe payload capacity(10¹¹ CFU/ml)
- Zero contamination
- 18 months stability
- 10x Smaller Logistics Footprint
- 100% water solubility

Reduce Chemical Intensity

- Transition to Sustainable Products Portfolio
- Bundle with your Plant Nutrition(Chemical) SKUs
- Enhance Nutrient Uptake & Nutrient Use Efficiency of Plants
- Maintain or Boost Yield
- FIB-SOL product treatments resulted in nominal yield increase with just 75% of Recommended Fertilizer Dosage in multiple crops

Proven Performance & Outcomes

- 6,00,000 acres since 2020
- Reliable Outcomes
- 10-25% yield boost across key crops
- FIB-SOL's products has cumulatively reached 6 lakh acres
- FY 2025-26: 1.85 Lakh units sold

Reduce Logistics Overhead

- 10X Lesser Volume
- 100ml SKU equivalent to 1 litre competitor products
- 5X lesser shelf/storage-space required

Avoid Return Risks

- Multi-season Stability
- Microbial encapsulation method keeps microbes dormant and viable for nearly 24 months
- Proven stability at higher temperature storage environments
- Claim 18 months shelf-life spanning 3 crop seasons

Easy Integration With Practice

- Thick-Liquid Form Product
- Farmers can apply products through:
Seed Dressing, Soil Drench, Drip Irrigation, Broadcasting, Foliar Spray

Soil Health Across Seasons

- Microbes enhance soil aggregation, nutrient recycling and water retention
- Microbes degrade soil carbon macromolecules into food for beneficial soil fauna
- Helps retain soil structure and resist drought conditions

Crop & Stage Specific Formulations

- Multiple crop and crop-stage specific microbial formulations
- Ensures best yield and quality outcomes

Flexible Manufacturing Services

- Agile Manufacturing Capability
- Only gel-based microbial formulation manufacturer globally
- MOQ flexibility
- Custom microbial compositions
- Deep vendor relationships for timely delivery

Elite Technical Support

- R&D heavy team of 10 members
- 4 Scientists
- 3 PhDs from IIT Madras
- 1 PhD from Madras University
- Network with faculty of top institutions
- Includes Biotechnologists, Biochemists & Chemists

Our Top Products

- Crop Nutrition
- Crop Protection
- Specials

Features of Product & Service - Marketing Companies

- IG4 Technology
- Novel Gel Carrier based formulation
- Industry-leading microbial payload(10^{13} CFU)
- Multi season stability(18 months)
- 100ml SKUs equivalent to 1 litre products
- Easy integration with farmer practices
- Reliable Yield & Quality Boost - 10-25% extra yield
- Sold across 1,85,000 acres (FY2025-26) & 6,00,000 acres cumulatively
- Agile Manufacturing Services
- In-house R&D Team lead by IIT Madras PhDs

Benefits of Our Products & Services

- Quickly add Advanced Microbials without R&D investments
- Transition Product Portfolio towards Sustainability
- Proven Reliable Product Performance
- Reduced Product Return Risk
- 5x Reduced Logistics Overheads
- 3x Reduced Shelf/Storage Space
- Extract 15-20% premium margins
- Flexible MOQs & Turn-around Time
- Quick Technical Support

Aggregators Pitch Manufacturing Service FAB

PRODUCTS & SERVICES

Your Ultimate Bio-input Partner

Patented Formulation Technology
IG4 Technology (Indian Patent No.: 421141)

Unique gel-based microbial carrier formulations deliver:

- Unmatched carrying capacity (10¹¹ CFU/ml)
- Multi-season stability (18 months)
- Microbial composition flexibility
- 100% water solubility

Reduce Chemical Intensity

- Biofertilizer organisms boost nutrient uptake
- Beneficial soil microbes fix Nitrogen, solubilize Phosphorus, mobilize Potassium
- Enhances nutrient uptake and nutrient use efficiency
- Field trials demonstrate nominal yield increase with 75% fertilizer dosage

Reduce Logistics Overhead

- 1000x More Microbes
- FCO spec: 10⁸ CFU/ml
- FIB-SOL products: 10¹¹ CFU/ml
- 10x lesser product volume requirement
- 5x lesser shelf/storage-space required

Easy Integration With Practice

- Thick-Liquid Form Product
- Seed Dressing
- Soil Drench
- Drip Irrigation
- Broadcasting
- Foliar Spray

Soil Health Across Seasons

- Better soil aggregation
- Nutrient recycling
- Water retention
- Soil structure retention
- Drought resistance

Crop & Stage Specific Formulations

- Multiple crop & stage specific formulations
- Better yield and quality outcomes

Proven Performance & Outcomes

- 6,00,000 acres since 2020
- 10-25% yield boost
- 100% Better quality produce
- Reduced diseases and pesticide requirements
- FY 2025-26: 1.85 lakh units sold

Flexible Manufacturing Services

- Agile manufacturing capability
- One and only gel-based microbial formulation manufacturer globally
- MOQ flexibility
- Custom microbial compositions
- Timely service delivery

Elite Technical Support

- R&D heavy team - 10 members
- 4 scientists

- 3 IIT Madras PhDs
- 1 Madras University PhD
- Includes Biotechnologists, Agronomists, Biochemists & Chemists

Features of Product & Service - Aggregators

- IG4 Technology
- Novel Gel Carrier based formulation
- Industry-leading microbial payload(10^{13} CFU)
- Multi season stability(18 months)
- 100ml SKUs equivalent to 1 litre products
- Easy integration with farmer practices
- Reliable Yield & Quality Boost - 10-25% extra yield
- Sold across 1,85,000 acres & 6,00,000 acres cumulatively
- Agile Manufacturing Services
- IIT Madras PhD led R&D Team

Benefits

- Add Advanced Microbials without R&D investments
- Sustainability transition
- Reduced Return Risk
- Reduced logistics overheads
- Premium margins
- Flexible MOQ
- Technical support

Products List Comprehensive

PRODUCT CATALOG

Products 1-55 include:

Sakthi Max, Surya Max, Super Soil, Super Tri, Super BMV, Super PAE, Super KMB, Super PSB, Super VAM, Tomato Magic, Chilli Magic, Onion Magic, Flowers Magic, Potato Magic, Veggies Magic, G2-Tomato, G2-Chilli, Banana Pro, Banana Max, G2-Onion, G2-Flowers, G2-Potato, G2-Veggies, Paddy Special, Citrus Special 1, Citrus Special 2, Turmeric Special 1, Turmeric Special 2, Maize Special 1, Maize Special 2, Cotton Special 1, Cotton Special 2, Groot Special, Ginger Special 1, Ginger Special 2, Sugarcane 1, Sugarcane 2, Tobacco Special, Nutrigel + Micronutrients, Nutri CN, Nutri KMS, Hortiking, Maize & Cereal Plus, Paddy Plus, BACI FORCE, BIO-SHIELD, VERTI-SHIELD, META-GUARD, Super Azoto, Super BT, The Garden, Biome, Decomposer, Mira, Mira Plus.

Core Organisms Mentioned:

- *Azospirillum lipoferum*
- *Bacillus megaterium*
- *Bacillus mucilaginosus*
- *Pseudomonas fluorescens*
- *Trichoderma viride*
- *Beauveria bassiana*
- *Metarrhizium anisopliae*
- *Verticillium lecanii*
- *Paecilomyces lilacinus*

- Vesicular arbuscular mycorrhiza (VAM)
- Azotobacter Chroococcum
- Rhizobium sp.
- Bacillus subtilis
- Bacillus thuringiensis
- Methanotroph sp.
- Pseudomonas stutzeri

Applications:

- Seed Treatment
- Soil Treatment
- Foliar Spray
- Seedling Treatment
- Drip Irrigation

Categories:

- Biofertilizer
- Biofungicide
- Bioinsecticide
- Bionematicide
- Micronutrients
- GHG Abatement

Major Modes of Action:

- Nitrogen fixation
- Phosphorus solubilization
- Potassium mobilization
- Root proliferation
- Systemic resistance induction
- Soil pathogen suppression
- Pest suppression
- Methane oxidation
- Nematode parasitism
- Rhizosphere enhancement

Key Benefits:

- Yield increase
- Better nutrient uptake
- Reduced pesticide dependency
- Reduced greenhouse gas emissions
- Better stress tolerance
- Improved soil health
- Sustainable farming support
- Compatible with Integrated Pest Management (IPM)

Detailed Mechanisms Included:

- Beauveria bassiana produces beauvericin and bassianolide toxins
- Metarrhizium anisopliae produces destruxins
- Bacillus subtilis produces iturin and surfactin
- Paecilomyces lilacinus secretes chitinases and proteases
- Methanotrophs oxidize methane and improve soil-air balance

Crop Specific Coverage:

- Paddy
- Citrus
- Turmeric
- Maize
- Cotton

- Ginger
- Sugarcane
- Tobacco
- Banana
- Tomato
- Chilli
- Onion
- Potato
- Flowers
- Vegetables

Additional Notes:

- Products support climate-smart agriculture
- Supports residue-free farming
- Improves drought tolerance
- Enhances crop vigor and resilience
- Supports sustainable disease management
- Enables long-term soil pest suppression
- Supports rhizosphere microbial balance